TU/CSE

TEZPUR UNIVERSITY

Spring Semester Mid Examination 2023

Course Code:CO525

Course Title: DATA MINING

Full Marks: 40

Time: *2Hrs*

(The figures in the right-hand margin indicate full marks for the questions)

1. Data Cleaning is a process that removes ____________ [1]
2. The output of KDD is ____________________ [1]
3. Interpret at least 4 major issues in data mining? [4]
4. How is unsupervised learning different from supervised learning? [4]
5. (a) What is data normalization?

   (b) Express the different techniques in handling the missing values? [2+4=6]
6. Compute the dissimilarity between objects of mixed attribute types given in Table below: [10]

Object identifier	test-1 (nominal)	test-2 (ordinal)	test-3 (numeric)	age	%fat in body	Calorie intake (Kcal)
1	code-A	excellent	45	23	26.5	1400
2	code-B	fair	66	27	7.8	1000
3	code-C	good	58	25	17.4	1200
4	code-A	excellent	88	23	27.2	1500

7. What are the two main disadvantages of k-means clustering? How can you overcome them? [5]

   OR

   Give the working of PAM algorithm.
8. Can you prioritize relative interconnectivity over relative closeness and vice versa? Justify with examples. [5]
9. Explain the working of BIRCH algorithm. [4]

Quick CO Mapping

Question No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
COs	C1	C1	C1	C1	C1	C1	C2	C2	C2

COs	Statements	Blooms Level
CO1	**Identify** the issues and challenges related to pre-processing and analysis of data to provide suitable input for data mining algorithms.	L2
CO2	**Analyze** various classification and clustering techniques and identify interesting patterns.	L4
CO3	**Use** pattern mining techniques to solve various market/sales correlations in large databases.	L3
CO4	Evaluate the performance of different data mining methods to **design** relevant applications considering a given dataset.	L6

TEZPUR UNIVERSITY TU/CSE

Class Test2, Spring 2023

Data Mining CO525

TOTAL MARKS: 25 **TIME: 45 mins**

1. Given d unique items, the total number of itemsets is 2^d. [1]

2. Apply FP-tree growth algorithm to generate frequent patterns for the database given below with min-sup=60%. Also show the FP-tree and the conditional FP-tree associated with each conditional node. [6+8= 14]

Transaction id	Items
T1	1,3,4
T2	2,3,5
T3	1,2,3,5
T4	2,5
T5	1,2,3,5

3. Give the difference of Apriori with FP tree growth algorithm. [4]
4. How can you overcome the disadvantage of Apriori algorithm? [6]

100

$$Info(D) = \sum_{i=1}^{m} P_i \log_2 P_i$$

$$info(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} info(D_j)$$

92.

32
9
14
~~55~~
30
85

33
15
48
9
57
+30
87

Entran → ~~80~~
30